Classical Fluorescence Denoising for Cell Segmentation

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I implemented and evaluated three classical restoration baselines in a team study of fluorescence image denoising. The study held Cellpose **cyto2** fixed to measure restoration through downstream instance segmentation. The team's archived comparison reports mean AP@0.5 of **0.5559 for BM3D**, versus **0.3171 for noisy input** [2].

1. My contribution

I adapted existing numerical libraries to the project's Poisson input format and evaluated the three outputs through the shared segmentation and scoring pipeline [1, 3].

Baseline	Implementation work	Source defaults
VST + BM3D	Count recovery, VST, noise scaling, inverse and batch I/O.	sigma_vst = 1.0
VST-TV	Normalized VST-domain Chambolle TV and safe inverse.	weight = 0.10 n_iter = 200
VST-Wiener	Local VST-domain filter, channel-wise I/O and safe inverse.	mysize = 5

I also contributed to the proposal's quantitative predictions, evaluation protocol, reproducibility plan and extension discussion, and to report/README preparation and review. The code uses **bm3d**, **scikit-image** and **SciPy**. Settings shown are pinned source defaults; the original benchmark launch configuration is not reconstructed here.

2. Evaluation design

The shared evaluation uses **68 Cellpose test image/mask pairs** with simulated Poisson noise. Per-image scales recover photon-count inputs for the VST. All methods use the same pretrained cyto2 segmenter, with **channels=[2, 1]** for green cytoplasm and red nucleus. AP@0.5 averages the per-image score below at IoU threshold 0.5 [1].

```
AP_i = TP_i / (TP_i + FP_i + FN_i)
```

3. Archived team results

Input to frozen segmenter	Mean AP@0.5	Role
Noisy images	0.3171	Reference
BM3D denoised images	**0.5559**	My classical baseline
VST-TV denoised images	0.5069	My classical baseline
VST-Wiener denoised images	0.4735	My classical baseline
Clean images	0.7343	Reference

All three baselines exceeded the noisy-input mean in this team comparison. BM3D ranked highest among the three, with an absolute gain of **0.2388 AP**. A gap of **0.1784 AP** remained to the clean-image reference. Values are archival team results, not new measurements.

Source links

[1] Pinned implementation: BM3D / TV / Wiener; segmentation / evaluation (revision e6b237d).
[2] Archived team benchmark figure. [3] My commits: BM3D (March 25); TV/Wiener (April 16).
[4] Selected team case figure; complete run guide and contribution record.

4. Selected cases and technical limits

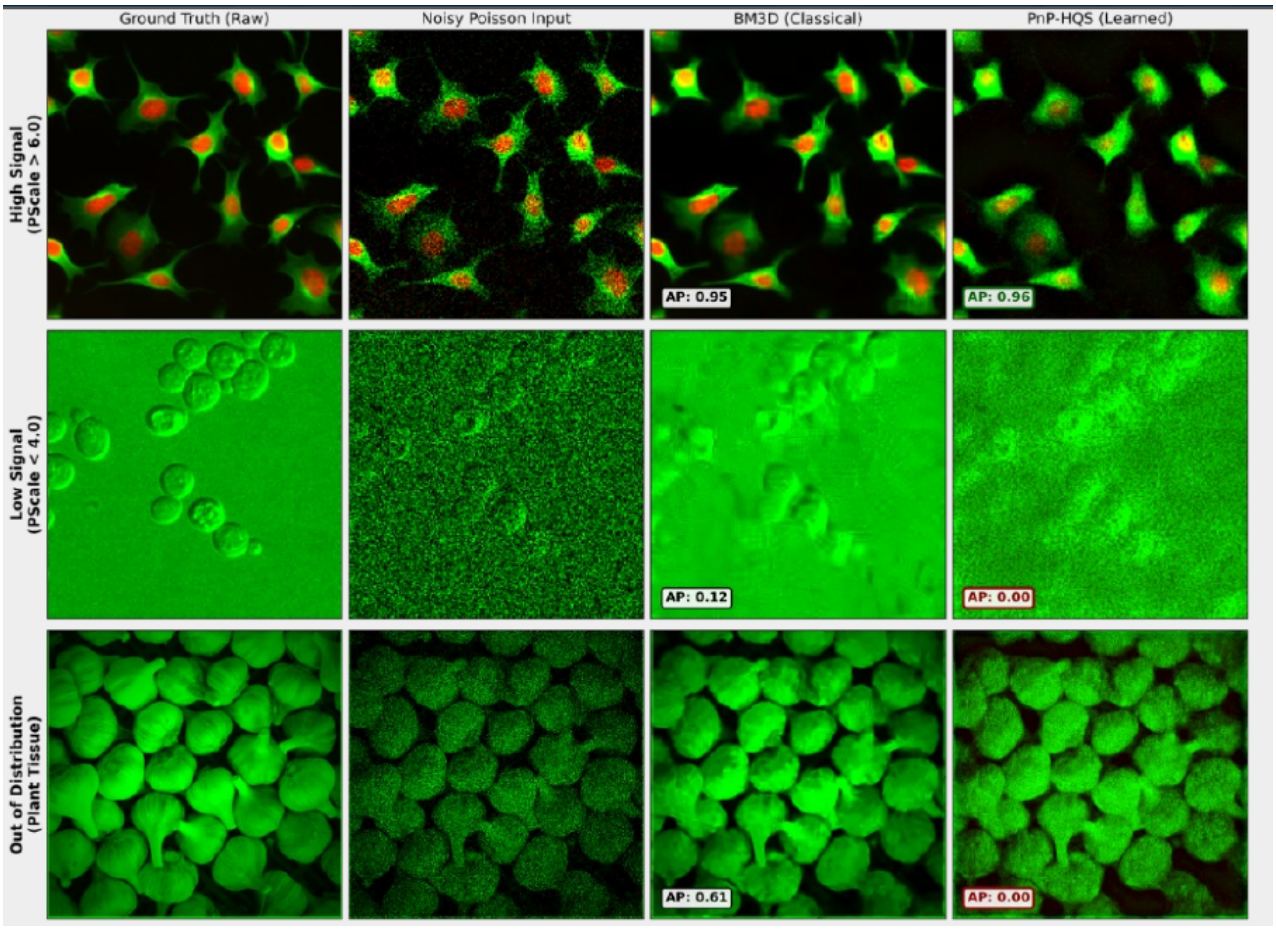


Figure 1. Team-selected cases from the EECS 556 presentation [4]. Columns are reference image, Poisson-corrupted input, BM3D and the team's PnP-HQS restoration. AP labels describe individual examples, separate from the 68-image means.

Top row, high-scoring example. Displayed AP is 0.95 for BM3D and 0.96 for PnP-HQS. Both perform well on this example; the small gap does not establish an overall ranking.

Middle row, difficult example. Displayed AP is 0.12 for BM3D and 0.00 for PnP-HQS. Visible noise suppression still leaves poor segmentation, motivating evaluation through the downstream task.

Bottom row, different method behavior. Displayed AP is 0.61 for BM3D and 0.00 for PnP-HQS, despite recognizable structures in both outputs. Appearance alone does not explain the score difference.

These selected cases do not establish a failure mechanism or a general claim about tissue types. Original sample identifiers, noise scales and masks were not independently checked for this summary.

5. Reproducibility and scope

No experiments were rerun to prepare this report. The linked run guide [4] follows fixed revision e6b237d, seed 42 and Cellpose 3.x cyto2. Most dependencies are unpinned, and the VST scripts use approximate inverse transforms. A full reproduction must preserve weights, versions and noise scales, and confirm the same 68 filenames across inputs, outputs and ground-truth masks; the evaluator can otherwise score an incomplete intersection.

Team and development credit. The project team was Ninad Kamath, Bingxu Lin and Ruoyu Wu; the upstream repository is owned by npkamath. Data preparation, segmentation, evaluation and other restoration/training extensions are team work. My implementation contribution covers the three classical baselines above. Development included my own coding and AI assistance.